MIL-C-15370/4B
5 June 1975
SUPERSEDING
MIL-C-15370/4A
29 December 1972

MILITARY SPECIFICATION SHEET

COUPLERS, DIRECTIONAL (WAVEGUIDE, MULTIHOLE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the coupler described herein shall consist of this document and the latest issue of Specification MIL-C-15370.

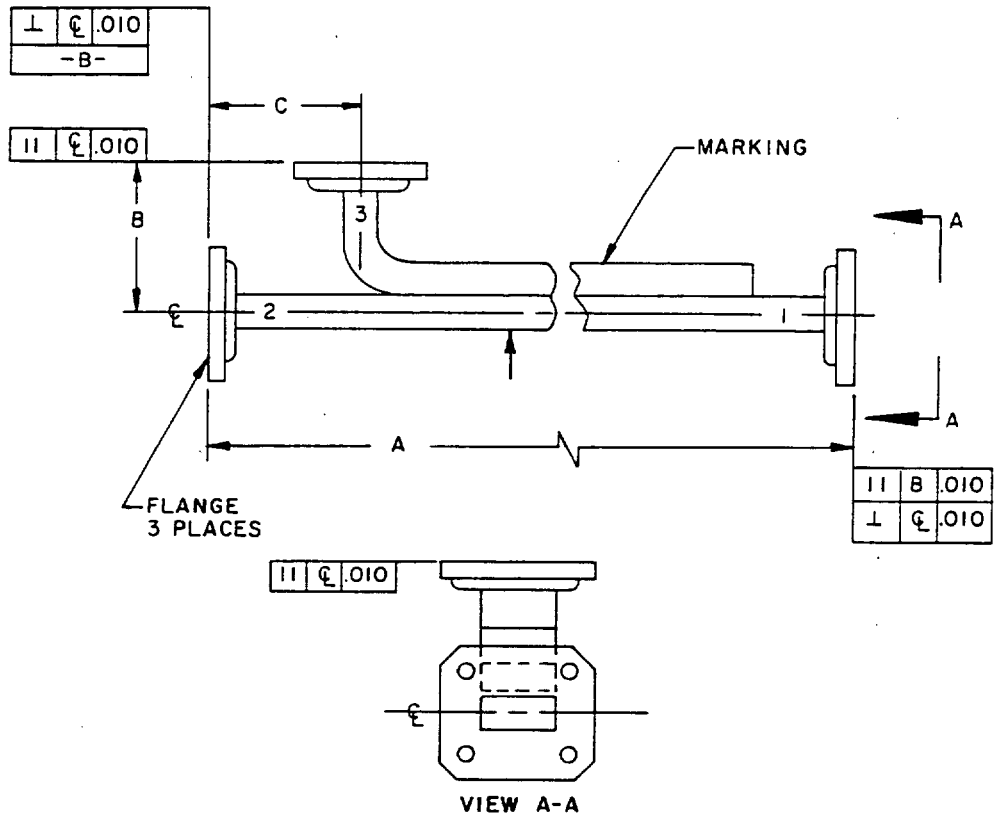


FIGURE 1. Dimensions and configuration.

FSC 5985

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:09Z
Check the source to verify that this is the current version before use.

TABLE I. Physical characteristics.

Dash number	Dimensions 1/ 2/ 3/					
	A		B		C	
	Min	Max	Min	Max	Min	Max
001	26.437 (671.50)	26.563 (674.70)	4.500 (114.30)	4.625 (118.48)	2.000 (50.80)	2.125 (53.98)
002	25.500 (647.70)	26.625 (676.28)	4.500 (114.30)	4.625 (118.48)	2.000 (50.80)	2.125 (53.98)
003	25.500 (647.70)	26.625 (676.28)	4.500 (114.30)	4.625 (118.48)	2.000 (50.80)	2.125 (53.98)
004	18.563 (471.50)	19.188 (487.38)	2.938 (74.63)	3.063 (77.80)	1.063 (27.00)	1.188 (30.18)
005	17.438 (442.90)	17.563 (446.10)	2.938 (74.63)	3.063 (77.80)	1.063 (27.00)	1.188 (30.18)
006	17.438 (442.90)	17.563 (446.10)	2.938 (74.63)	3.063 (77.80)	1.063 (27.00)	1.188 (30.18)
007	16.625 (422.28)	16.750 (425.45)	2.938 (74.63)	3.063 (77.80)	1.000 (25.40)	1.125 (28.58)
008	15.625 (396.88)	15.750 (400.05)	2.938 (74.63)	3.063 (77.80)	1.000 (25.40)	1.125 (28.58)
009	15.625 (396.88)	15.750 (400.05)	2.938 (74.63)	3.063 (77.80)	1.000 (25.40)	1.125 (28.58)
010	12.188 (309.58)	12.313 (312.75)	1.968 (50.00)	2.094 (52.79)	0.687 (17.48)	0.813 (20.68)
011	9.313 (236.55)	10.656 (270.66)	1.937 (49.20)	3.000 (76.20)	0.562 (14.27)	1.000 (25.40)
012	9.313 (236.55)	9.968 (253.19)	1.937 (49.20)	3.000 (76.20)	0.562 (14.27)	1.000 (25.40)
013	7.562 (192.10)	9.968 (253.19)	1.937 (49.20)	3.000 (76.20)	0.562 (14.27)	1.000 (25.40)
014	20.350 (516.89)	20.410 (518.41)	2.230 (56.64)	2.270 (57.66)	0.980 (24.89)	1.020 (25.91)
015	20.350 (516.89)	20.410 (518.41)	2.230 (56.64)	2.270 (57.66)	0.980 (24.89)	1.020 (25.91)
016	20.350 (516.89)	20.410 (518.41)	2.230 (56.64)	2.270 (57.66)	0.980 (24.89)	1.020 (25.91)
017	20.350 (516.89)	20.410 (518.41)	2.230 (56.64)	2.270 (57.66)	0.980 (24.89)	1.020 (25.91)
018	5.975 (151.77)	8.750 (222.25)	1.630 (41.40)	1.938 (49.23)	0.400 (10.16)	0.719 (18.26)
019	5.975 (151.77)	8.750 (222.25)	1.630 (41.40)	1.938 (49.23)	0.400 (10.16)	0.719 (18.26)

1/ Dimensions are in inches.
2/ Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3/ Metric equivalents are in parentheses.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:09Z
Check the source to verify that this is the current version before use.

TABLE II. Electrical characteristics.

Dash number	Frequency range	Coupling factor	Mean coupling accuracy 1/	Coupling variation 1/	Effective directivity (min) 1/	VSWR (max)		Power dissipation of internal load (avg)	Equivalent waveguide	Flange
						Primary line	Secondary line			
	GHz	dB	dB	dB	dB			watts		
001	5.85- 8.2	3	±0.5	±0.5	40	1.10	1.20	1	M85/1-067 (RG-50/U)	M3922/55-001 (UG-344/U)
002	5.85- 8.2	10	±0.5	±0.5	40	1.05	1.15	1	M85/1-067 (RG-50/U	M3922/55-001 (UG-344/U)
003	5.85- 8.2	20	±0.5	±0.5	40	1.05	1.15	1	M85/1-067 (RG-50/U)	M3922/55-001 (UG-344/U)
004	7.05-10.0	3	±0.5	±0.5	40	1.10	1.20	1	M85/1-073 (RG-51/U)	M3922/53-002 (UG-51/U)
005	7.05-10.0	10	±0.5	±0.5	40	1.05	1.15	1	M85/1-073 (RG-51/U)	M3922/53-002 (UG-51/U)
006	7.05-10.0	20	±0.5	±0.5	40	1.05	1.15	1	M85/1-073 (RG-51/U)	M3922/53-002 (UG-51/U)
007	8.2 -12.4	3	±0.5	±0.5	40	1.10	1.20	1	M85/1-079 (RG-52/U)	M3922/53-001 (UG-39/U)
008	8.2 -12.4	10	±0.5	±0.5	40	1.05	1.15	1	M85/1-079 (RG-52/U)	M3922/55-001 (UG-39/U)
009	8.2 -12.4	20	±0.5	±0.5	40	1.05	1.15	1	M85/1-079 (RG-52/U)	M3922/53-001 (UG-39/U)
010	12.4 -18.0	20	±0.5	±0.5	40	1.05	1.15	1	M85/1-089 (RG-91/U)	M3922/53-005 (UG-419/U)
011	18.0 -26.5	3	±0.5	±0.5	40	1.10	1.20	0.5	M85/1-106 (RG-66/U)	M3922/54-001 (UG-595/U)
012	18.0 -26.5	10	±0.5	±0.5	40	1.05	1.15	0.5	M85/1-106 (RG-66/U)	M3922/54-001 (UG-595/U)
013	18.0 -26.5	20	±0.5	±0.5	40	1.05	1.15	0.5	M85/1-106 (RG-66/U)	M3922/54-001 (UG-595/U)
014	11.0 -17.0	10	±0.3	0.4	35	1.10	1.20	1	M85/1-090 (RG-349/U)	UG-1665 2/
015	11.0 -17.0	15	±0.3	0.4 3/	35	1.10	1.20	1	M85/1-090 (RG-349/U)	M3922/53-012

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:09Z
Check the source to verify that this is the current version before use.

TABLE II. Electrical characteristics - Continued.

Dash number	Frequency range	Coupling factor	Mean coupling accuracy 1/	Coupling variation 1/	Effective directivity (min) 1/	VSWR (max)		Power dissipation of internal load (avg)	Equivalent waveguide	Flange
						Primary line	Secondary line			
	GHz	dB	dB	dB	dB			watts		
016	11.0 -17.0	20	±0.3	0.4 3/	35	1.10	1.20	1	M85/1-090 (RG-349/U)	M3922/53-012
017	11.0 -17.0	30	±0.3	0.4 3/	35	1.10	1.20	1	M85/1-090 (RG-349/U)	M3922/53-012
018	26.5 -40.0	20	±0.7	±0.6	40	1.05	1.20	0.5	M85/3-008 (RG-271/U)	M3922/54-003 (UG-599/U)
019	26.5 -40.0	30	±0.7	±0.6	40	1.05	1.20	0.5	M85/3-006 (RG-96/U)	M3922/54-003 (UG-599/U)

1/ Measured by swept frequency techniques.
2/ Except the lip thickness is .25" and tapped holes .138-32UNC-2B
3/ Total change across the frequency band.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:09Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Design, construction, and physical dimensions: See figure 1 and table I.

Equivalent waveguide: See table II.

Flange: See table II.

Operating temperature range: -40°C to +85°C for dash numbers -001 thru -017.
-20°C to +65°C for dash numbers -018 and -019.

Frequency range: See table II.

Coupling: See table II.

Mean coupling: See table II.

Coupling variation: See table II.

Effective directivity: See table II.

VSWR: See table II.

Marking: See figure.

Part number: M15370/4-(and a dash number from table I).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:09Z
Check the source to verify that this is the current version before use.

TABLE III. Cross reference of part number/AN nomenclature.

Dash number	AN nomenclature
001	CU-1510/U
002	CU-1511/U
003	CU-1512/U
004	CU-1507/U
005	CU-1508/U
006	CU-1509/U
007	CU-1513/U
008	CU-1514/U
009	CU-1515/U
010	CU-1519/U
011	CU-1522/U
012	CU-1523/U
013	CU-1524/U

Note: Directional couplers covered by this specification sheet are substitutable for the commercial types listed in table IV.

TABLE IV. Substitution data.

Part number M15370/4-	FSN 5985-	Manufacturer's code	Commercial type
001	679-0637	16055	C201A
001	679-0637	28480	J752A
002	679-0638	28480	J752C
003	679-0639	16055	C201D
003	679-0639	28480	J752D
004		28480	H752A
005		28480	H752C
006		28480	H752D
007	838-9920	28480	X752A
007	838-9920	00341	DD174X1-3DB
008	646-9472	94117	1292022
008	646-9472	28480	X752C
009	679-0645	19200	7904840
009	679-0645	80006	414796
009	679-0645	28480	X752D
010	562-8870	28480	P752D
011		00929	K610A
012	692-8714	00929	K610C
013		00929	K610D
014		05869	712402-1
014		00443	70467P1
015		05869	712402-2
015		00443	70467P2
016		05869	712402-3
016		00443	70467P3
017		05869	712402-4
017		00443	70467P4
018	084-8991	28480	R752D
018		94990	58-P06571D
019		34228	885C052581
019		96341	MA-772C

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:09Z
Check the source to verify that this is the current version before use.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - SG
Navy -
Air Force - 11, 17, 80
DSA - ES

User activities:
Army - WC
Navy - AS, CG, MC, SH, OS
Air Force - None

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5985-0785)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:09Z
Check the source to verify that this is the current version before use.